Tape-Spring Boom L001 [0/0/0]: Axial, Opposite-Sense Bending and Coiling Checks

ME295 thin-shell project (Benazzo, Venkatadri) | 11 November 2026 | OpenSeesPy shell models built with the project skills in `.claude/skills`

Summary

A 1 m long composite tape spring (IM7/8552 carbon/epoxy, catalog material L001, three 0 deg plies, transverse radius 20 mm, 180 deg arc) clamped at one end was checked for three load cases: a 10 N axial pull, a 1 N transverse tip load in opposite-sense bending, and coiling around a 40 mm radius hub. For each case the ply stresses were checked with the 2D Hashin failure criterion, where failure is predicted at F = 1. The strength ratio R is the factor by which the load (or the stresses) could be multiplied before F reaches 1. **No case is close to material failure.** Coiling is the governing case (F = 0.18, R = 2.3). The boom mass is **10.42 g**.

Load case	Deformation	Max Hashin F	Strength ratio R	Critical mode / location
Axial pull, 10 N	tip extension 0.0096 mm	4.0×10^{-7}	1576	fibre tension, uniform along the boom
Tip load 1 N, opposite-sense bending	**tip deflection 8.52 mm**	4.2×10^{-4}	49	fibre tension, root, tape edges
Coiling on 40 mm hub	flattened (k_y = 1/20 mm^{-1}) and bent (k_x = 1/40 mm^{-1})	**0.183**	2.34	matrix tension, outer ply

1. The boom and its material

Geometry. Circular-arc cross-section with radius R = 20 mm and subtended angle 180 deg, so the flattened width is b = pi R = 62.83 mm. Length L = 1000 mm along x. The arc is symmetric about y = 0 with its centre of curvature on +z: the crown is at z = 0 and the two free edges at z = 20 mm. The centroid of the arc is at z = R(1 - 2/pi) = 7.27 mm and its second moment of area about the horizontal centroidal axis is $I = R^3 h (pi/2 - 4/pi) = 250\ mm^4$.

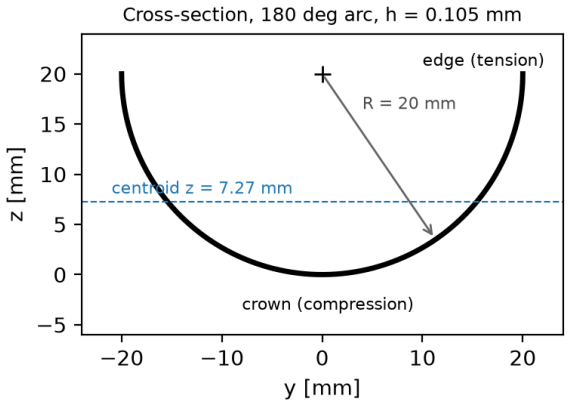


Figure 1. Cross-section. In opposite-sense bending the edges are in tension and the crown in compression.

Material and layup. L001 in the project UD lamina catalog is Hexcel IM7/8552 (NCAMP, room temperature dry mean values). The project uses a ply thickness of 0.035 mm for every catalog row, so the [0/0/0] laminate is h = 0.105 mm thick, with all fibres along the boom axis.

E_1	E_2	G_{12}	nu_{12}	X_t	X_c	Y_t	Y_c	S_{12}	t_{ply}	rho
158.5 GPa	8.95 GPa	4.67 GPa	0.316	2501 MPa	1716 MPa	64 MPa	285.7 MPa	91.1 MPa	0.035 mm	1580 kg/m^3

The laminate-properties skill gives the classical lamination theory (CLT) stiffness: A_{11} = 16737, A_{22} = 945, A_{12} = 299, A_{66} = 490 N/mm; B = 0 (symmetric); D_{11} = 15.38, D_{22} = 0.868, D_{12} = 0.274, D_{66} = 0.451 N mm. The same skill measured the stiffness the OpenSeesPy section actually has, by imposing strains on one element: A and B agree with CLT exactly and D is 0.4% lower, the expected effect of the layer integration. The axial modulus of the laminate is $E_x = (A_{11} - A_{12}^2/A_{22})/h$ = 158.5 GPa, so the bending stiffness of the boom is EI = 3.96 x 10^7 N mm^2.

2. How the load cases were evaluated

Finite-element model. Each case was written as a case file and turned into one standalone OpenSeesPy script by the project skills: *laminate-properties* (orthotropic ply material and LayeredShell section), *shell-mesh* (the arc extruded along x and meshed with 4-node shells), *shell-bc-load* (supports, loads, solver, results file) and *shell-assemble-run*, which joins the three code pieces, checks that they fit, and runs the script. The mesh is 400 x 25 ASDShellQ4 elements of 2.5 x 2.5 mm (10,000 elements, 62,562 degrees of freedom). The root edge x = 0 is fully clamped. The tip edge x = L is attached to a rigid end fixture whose master node carries the load, which is how the project loads curved thin sections; a free arc edge would be about 8% too flexible. The solutions are linear static, solved with 16 CPU threads.

Load cases. (1) Axial: F_x = +10 N on the tip fixture. (2) Bending: F_z = -1 N on the tip fixture. With the crown at the bottom, a downward tip load puts the edges in tension and gives a longitudinal curvature opposite in sign to the transverse curvature: this is *opposite-sense* bending (the project convention, shell-buckling sign +1). (3) Coiling: there is no FE run. The ply stresses of a flange that is flattened and wound on the hub follow from CLT alone, as described below.

Failure check. The *failure-check* skill reads the membrane strains and curvatures of every element at its four Gauss points and computes the ply stresses at the bottom, middle and top of each ply. These are exact CLT kinematics, then the ply stiffness. It then evaluates the 2D Hashin criterion (Benazzo & Pellegrino, AIAA SciTech 2025, Eq. 3). Only the mode matching the stress sign is active:

fibre tension (s_1 >= 0)	$F = (s_1/X_t)^2 + (t_{12}/S_{12})^2$
fibre compression (s_1 < 0)	$F = (s_1/X_c)^2$
matrix tension (s_2 >= 0)	$F = (s_2/Y_t)^2 + (t_{12}/S_{12})^2$
matrix compression (s_2 < 0)	$F = (s_2/Y_c)^2 + (t_{12}/S_{12})^2$ (with the default $S_{23} = Y_c/2$)

The largest F over all elements, Gauss points, plies and through-thickness points is reported with its strength ratio R. Because the analyses are linear, R is also a load factor: the load could be multiplied by R before F reaches 1.

Coiling. A tape spring wound on a hub is flattened transversely (curvature change $k_y = 1/R$ = 0.05 mm^{-1}) and bent along its length to the hub radius ($k_x = 1/R_{hub}$ = 0.025 mm^{-1}). The failure-check command-line tool covers only the flattening. The hub bending was added in a short script (`coil_hub_check.py`) that calls the skill's own stress and Hashin functions with (k_x, k_y) and zero membrane force. Which face ends up outside depends on how the tape is wound, so all four sign combinations of k_x and k_y were checked and the largest F kept. The hub radius 40 mm was used as the coil radius of the mid-surface. The true value is 40.05 mm, so this is slightly conservative.

Mass. The *objective-constraints* skill sums the undeformed element areas times the areal density rho h, with rho = 1580 kg/m^3 taken from the L001 catalog row.

Checks. Every result was compared with a hand calculation (bar or beam theory). The bending case was rerun on a 1.25 mm mesh (40,000 elements). Whether the bending stays in the linear regime was checked separately with the *shell-buckling* skill (section 3.2).

3. Results

3.1 Axial pull, 10 N

The tip moves **0.00957 mm** along the axis, against PL/(E$_x$ h b) = 0.00956 mm from bar theory (+0.06%). The axial stress in the fibres is about 1.5 MPa everywhere: 10 N / (62.83 x 0.105) mm^2 = 1.52 MPa, rising to 1.59 MPa next to the clamp, where the clamp restrains Poisson contraction. The Hashin index is F = 4.0 x 10^{-7} (fibre tension), a strength ratio of R = 1576. Material failure in tension would need a load of about 15.8 kN.

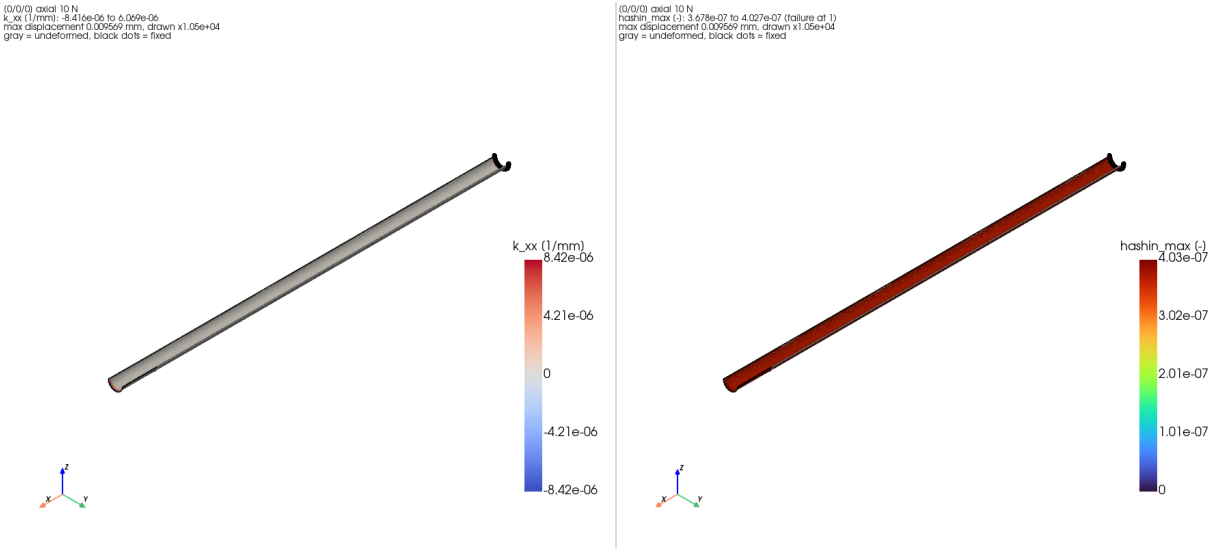


Figure 2. Axial 10 N: deformed shape drawn x10,500 over the undeformed outline (gray). Left: longitudinal curvature k$_{xx}$, essentially zero. Right: Hashin index, uniform at 4 x 10^{-7}.

3.2 Transverse tip load, 1 N, opposite-sense bending

The tip deflects **8.52 mm** downward and rotates 0.0127 rad. Cantilever beam theory, PL^3/(3EI) with EI = 3.96 x 10^7 N mm^2, gives 8.41 mm. The shell model is 1.3% more flexible, which is expected because the clamped and loaded ends of a thin curved section are not perfectly rigid. Halving the element size to 1.25 mm changes the deflection to 8.50 mm (-0.2%), so the mesh is converged.

The largest stress is at the root, where the bending moment is P L = 1000 N mm. It is fibre tension at the two free edges: s$_1$ = 50.0 MPa, against M c/I = 1000 x 12.73 / 250 = 50.9 MPa from beam theory, with c = 20 - 7.27 = 12.73 mm. The crown is in compression. The Hashin index is **F = 4.2 x 10^{-4}** (fibre tension), a strength ratio of R = 49. The fibres carry the bending stress and are very strong, so material failure is far away.

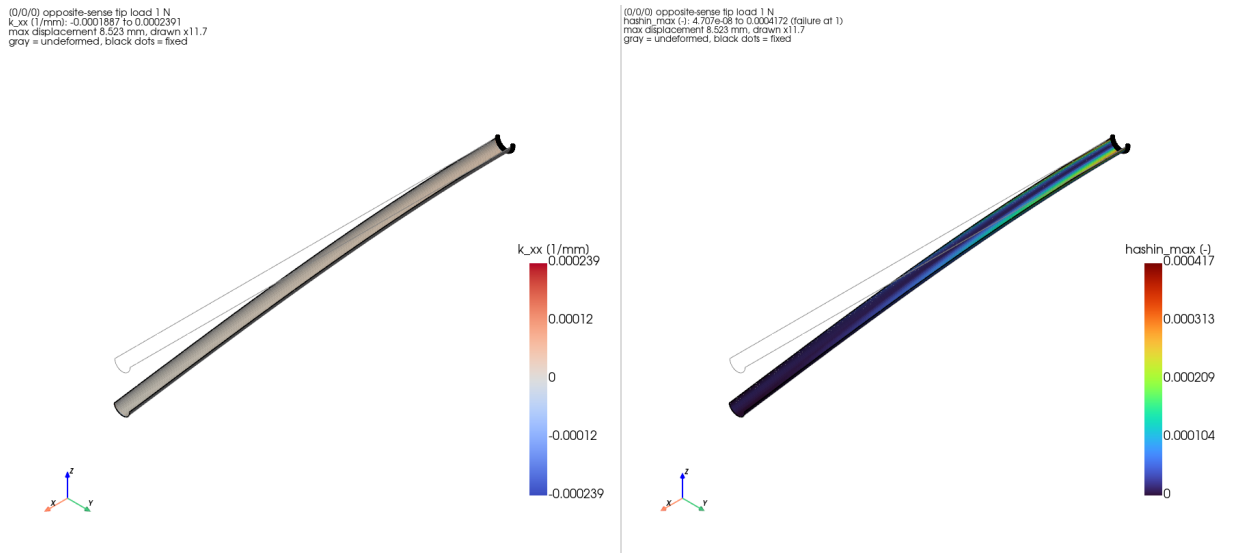


Figure 3. Tip load 1 N, opposite-sense: deformed shape drawn x11.7. Left: longitudinal curvature k_{xx}. Right: Hashin index, largest along the root edges.

Is the linear analysis valid? At the scale of the whole beam, yes: $PL^2/EI = 0.025$, far below the value of about 1 where large-rotation effects matter. A tape spring can also snap when its cross-section flattens. Under opposite-sense bending this is a limit point, which a linear analysis cannot see. To check it, a 200 mm segment under a uniform opposite-sense end moment was followed with rotation control (shell-buckling skill, corotational ASDShellQ4, phase B only, perfect geometry). The moment-rotation path stays linear (197,000 vs EI/L = 198,000 N mm/rad). It was still rising at 3,328 N mm when it was stopped, more than 3.3 times the 1000 N mm root moment of this load case (Figure 4). The 1 N load therefore causes no snap-through, and the linear result stands. The actual snap moment was not computed. Two nonlinear runs of the full cantilever (load and displacement control) stopped converging near 0.25 N while their paths were still exactly linear. This was a solver problem with the rigid tip fixture at large rotations, not a structural limit point.

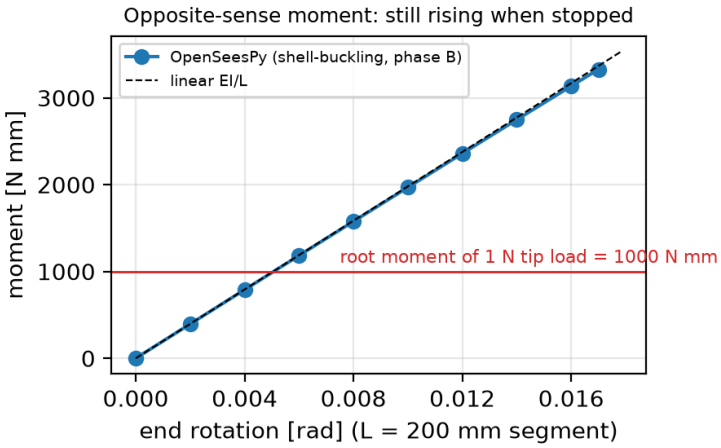


Figure 4. Opposite-sense moment vs end rotation of a 200 mm segment (OpenSeesPy, 2.5 mm mesh) against linear theory.

3.3 Coiling around a 40 mm radius hub

Flattening alone (failure-check skill) gives F = 0.136. With the hub curvature added, the worst of the four sign combinations is **F = 0.183** (matrix tension, strength ratio R = 2.34). It occurs on the outer face of the outer ply when the flattening and the coiling curvature have the same sign: there, s_1 = 217 MPa (fibre tension, F

contribution 0.008) and s_2 = 27.4 MPa transverse. The transverse stress governs because Y_t = 64 MPa is 40 times lower than X_t. The surface strains are h/(2R) = 0.26% across the boom and h/(2R_{hub}) = 0.13% along it. Coiling is the governing load case of this boom, but its margin is still comfortable.

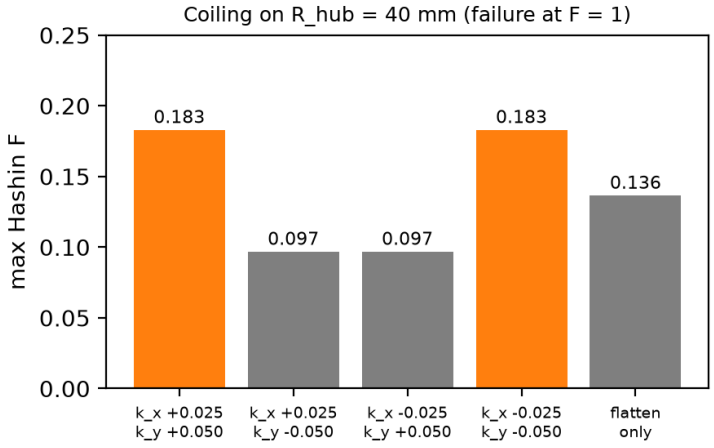


Figure 5. Hashin index of the coiled flange for the four curvature sign combinations (k in mm^{-1}) and for flattening only.

4. Mass

The objective-constraints skill gives a boom mass of **10.42 g** (0.01042 kg), or 10.4 g/m. A hand check gives the same value: rho b h L = 1580 kg/m^3 x 62.83 mm x 0.105 mm x 1000 mm = 10.42 g. The rigid end fixtures used to apply the loads are not part of the structure and have no mass.

5. Assumptions and limitations

- Strengths are catalog mean values (not design allowables) measured on 0.18 mm plies. Thin 35 um plies usually have higher in-situ transverse and shear strengths, so the matrix-mode indices are probably conservative. S_{23} is not in the catalog; the default S_{23} = Y_c/2 was used. It only affects matrix compression, which does not govern any case here.

- The coiling check is the uniform, fully coiled state. The transition zone, where the section opens from flat back to its arc shape, has extra stresses this check does not include, so some margin is needed. The coiling F of 0.18 leaves plenty. Stress relaxation during long stowage is not modelled.

- The bending result is linear; the snap check shows that the 1 N load is well below the opposite-sense snap moment, but the snap moment itself and equal-sense bending were not evaluated.

- Stresses next to the clamp corners are mesh dependent. Here the peak bending stress matches beam theory within 2%, so it is not a corner singularity.

6. Files

All in `Code for Project/11-11-26 (Tara)/`: case files `ts_L001_axial.json`, `ts_L001_opp_bending.json` (and the 1.25 mm mesh and nonlinear variants); the assembled OpenSeesPy scripts `*_opensees.py`; laminate `L001_000.json`; coiling check `coil_hub_check.py`; snap check `ts_L001_opp_moment_buckling.json`; results, Hashin reports, mass JSON and logs in `results/`; figures in `report_figs/`; this report from `build_report.py`.